

2016.0 RANGE ROVER (LG), 206-00

BRAKE SYSTEM - GENERAL INFORMATION

BRAKE SYSTEM PRESSURE BLEEDING

[G1549825]

GENERAL PROCEDURES

70.25.06	BRAKES - HIGH PRESSURE SYSTEM - BLEED	ALL DERIVATIVES	0.5	USED WITHINS
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CHECK

WARNING:

If any components upstream of the Hydraulic Control Unit (HCU), including the HCU itself are replaced, the brake system must be bled using Land Rover approved diagnostic equipment. This will ensure that all air is expelled from the new component(s).

CAUTIONS:

- Brake fluid will damage paint finished surfaces. If spilled, immediately remove the fluid and clean the area with water.
- The brake fluid reservoir must remain full with new, clean brake fluid at all times during bleeding.

NOTE:

Bleeding of the complete brake system must be carried out using Land Rover approved diagnostic equipment. Where only the primary or secondary brake circuits have been disturbed in isolation, it should only be necessary to bleed that circuit. Partial bleeding of the hydraulic system is only permissible if a brake tube or hose has been disconnected with only minimal loss of fluid.

1.

WARNING:

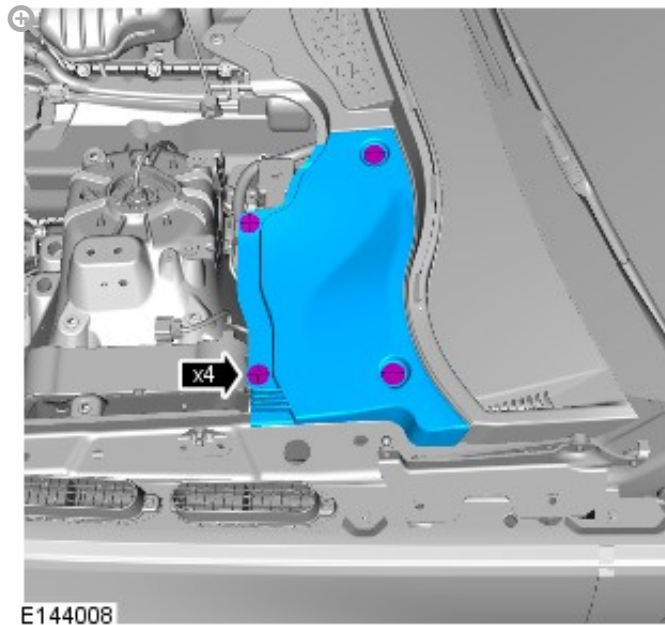
Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

2.

NOTE:

LHD illustration shown, RHD is similar.



3. Check that the brake fluid lines are secure and that there are no signs of a brake fluid leak. If a brake fluid leak is detected, investigate and rectify the cause of the leak before bleeding the brakes.

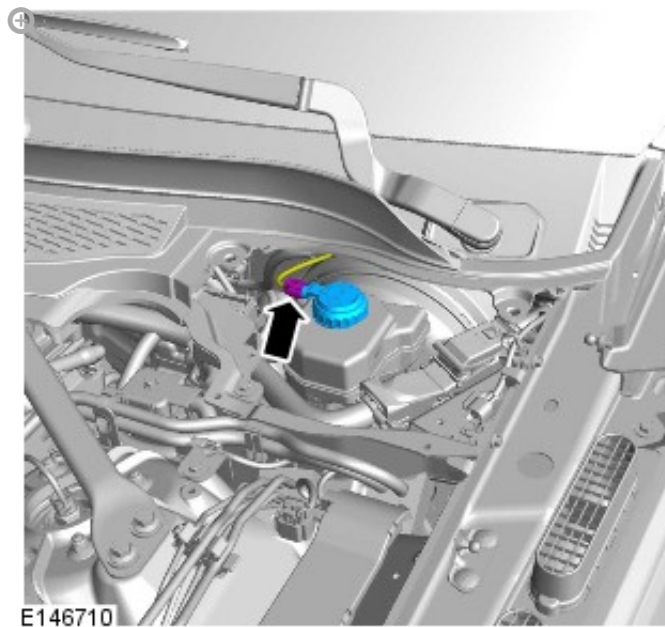
4.

WARNING:

Do not allow dirt or foreign liquids to enter the reservoir. Use only new brake fluid of the correct specification from airtight containers. Do not mix brands of brake fluid as they may not be compatible.

NOTE:

LHD illustration shown, RHD is similar.



Fill the brake fluid reservoir to the MAX mark.

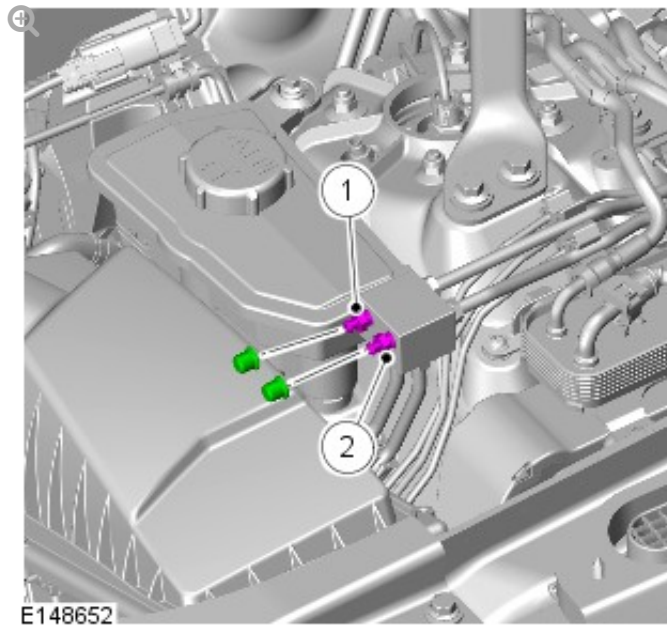
5. Conduct the bleed procedure with the engine running.

ADJUSTMENT

6.

NOTES:

- This step only applies if the vehicle is fitted with brake bleed block.
- This step must be carried out only if the Anti-Lock Brake System (ABS) module or the Master cylinder have been removed from the vehicle.
- LHD illustration shown, RHD is similar.



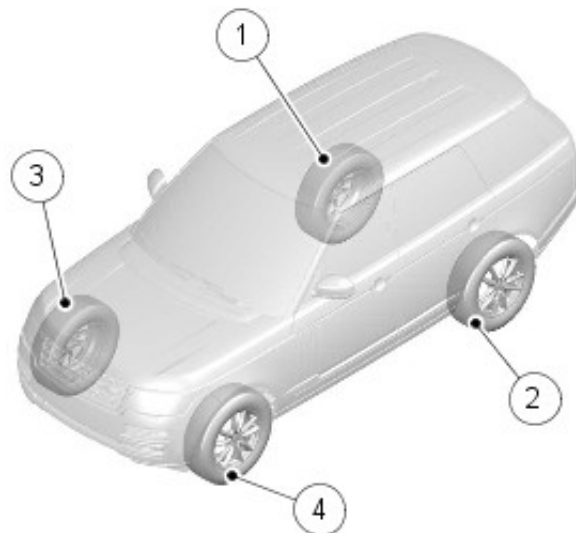
1. Install the bleed tube to the bleed screw and immerse the free end of the bleed tube in a bleed jar containing a small quantity of approved brake fluid.
2. With assistance, depress the brake pedal steadily through its full stroke and allow it to return to the rest position. Repeat the procedure until brake fluid, clean and air-free flows into the bleed jar.
3. When brake fluid, clean and air-free flows into the bleed jar, depress and hold the brake pedal down. Tighten the bleed screw.

Torque: 18 Nm

7. Fill the brake fluid reservoir to the MAX mark.

Left-hand drive vehicles

8.

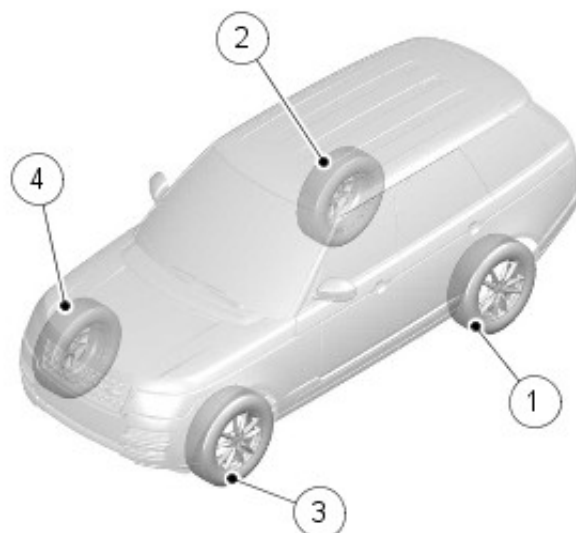


E146994

Starting at the brake caliper furthest away from the brake master cylinder, loosen the bleed screw by one-half to three-quarters of a turn.

Right-hand drive vehicles

9.



E146995

All vehicles

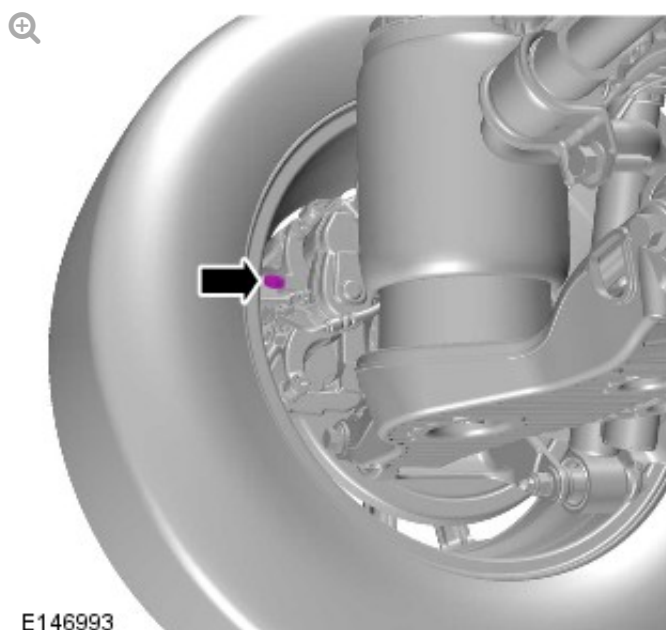
10. Install the bleed tube to the brake caliper bleed screw and immerse the free end of the bleed tube in a bleed jar containing a small quantity of approved brake fluid.

11. With assistance, depress the brake pedal steadily through to 2/3 of its full stroke.
12. With the brake pedal held down, close bleed screw and then return the brake pedal to 1/3 of its full stroke and hold.
13. Repeat steps 3 to 5, 28 times for the rear brake and 10 times for the front brake.
14. At the end of the bleed process, depress and hold the brake pedal down.

15.

CAUTION:

Make sure the bleed screw cap is installed after bleeding.
This will prevent corrosion to the bleed screw.



With the brake pedal fully depressed, tighten the bleed screw.

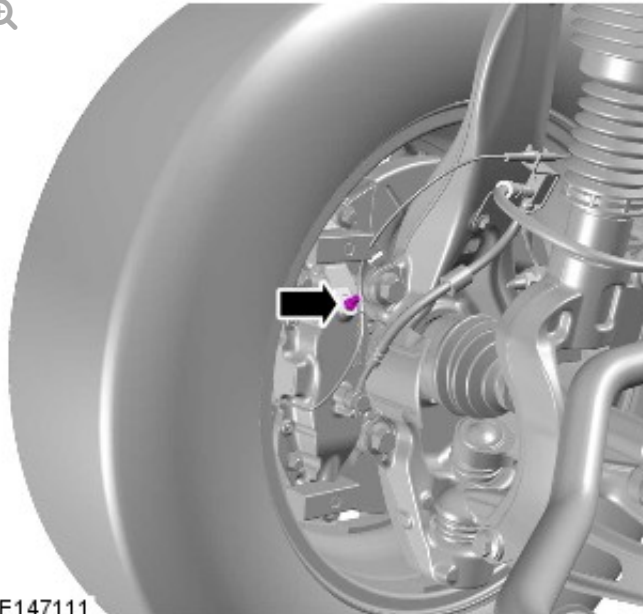
Torque: 11 Nm

Vehicles with standard brakes

16.

CAUTION:

Make sure the bleed screw cap is installed after bleeding.
This will prevent corrosion to the bleed screw.



E147111

With the brake pedal fully depressed, tighten the bleed screw.

Torque: 8 Nm

Vehicles with high performance brakes

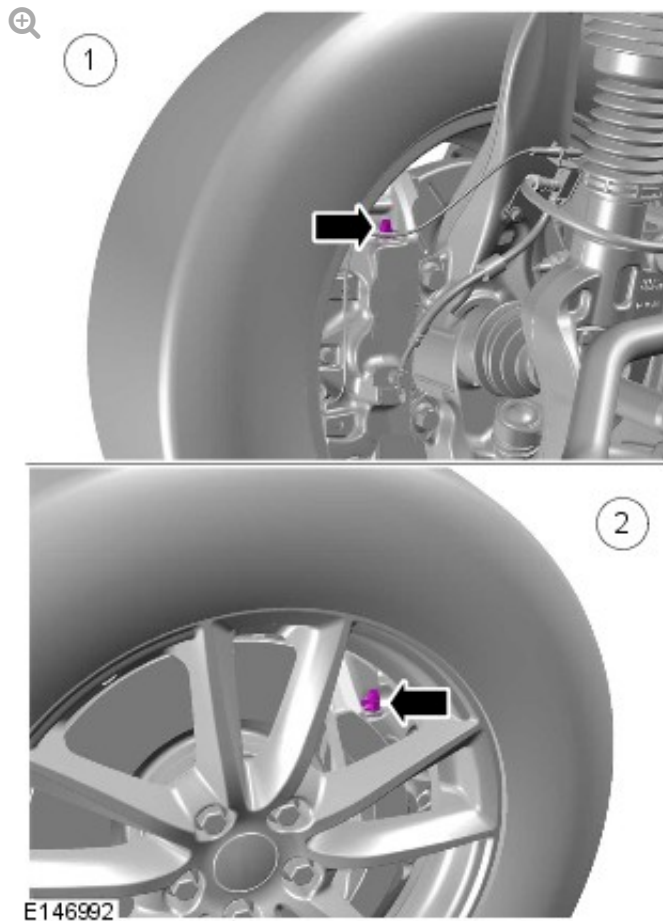
17.

WARNING:

High performance front brake calipers have two bleed points.
The inner bleed point must be bled before the outer bleed point. Braking efficiency may be seriously impaired if an incorrect bleed sequence is used.

CAUTION:

Make sure the bleed screw cap is installed after bleeding.
This will prevent corrosion to the bleed screw.



With the brake pedal fully depressed, tighten the bleed screw.

Torque: 8 Nm

All vehicles

18. Fill the brake fluid reservoir to the MAX mark.

19.

WARNING:

Braking efficiency may be seriously impaired if an incorrect bleed sequence is used.

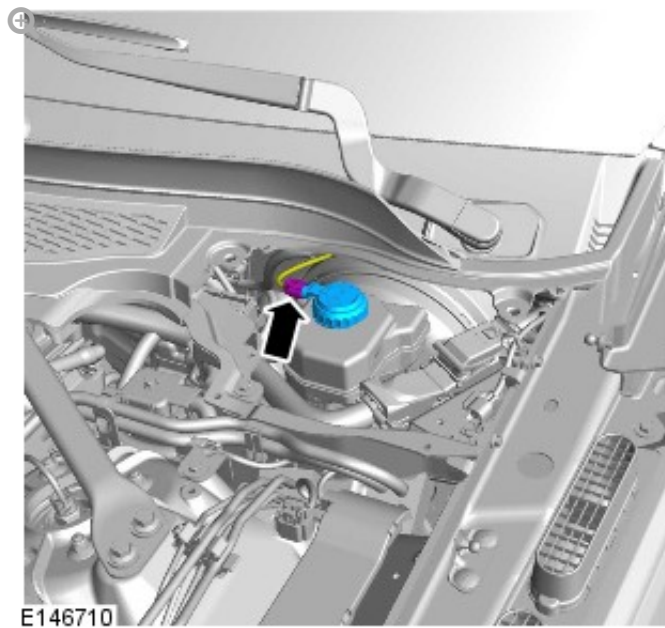
Repeat the brake bleeding procedure for each brake caliper, following the above sequence.

20. Apply the brakes and check for leaks.

21.

NOTE:

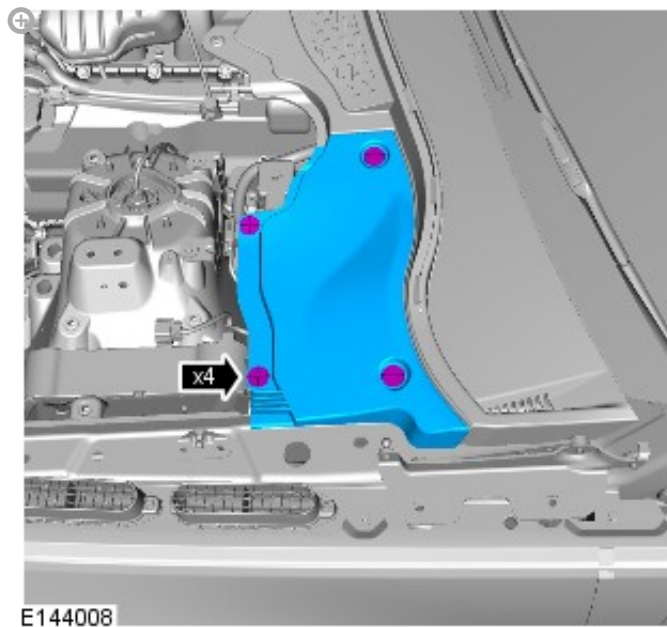
LHD illustration shown, RHD is similar.



22.

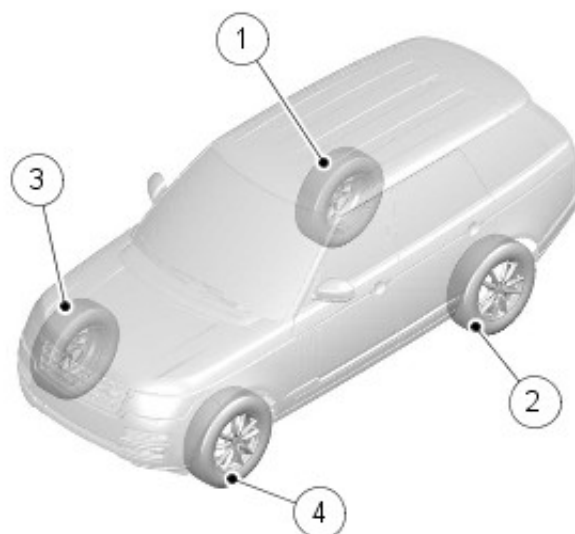
NOTE:

LHD illustration shown, RHD is similar.



i Left-hand drive vehicles

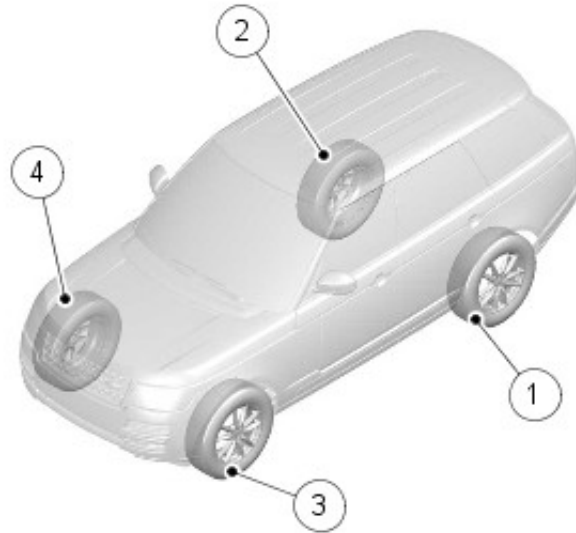
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E146994

Starting at the brake caliper furthest away from the brake master cylinder, loosen the bleed screw by one-half to three-quarters of a turn.

i Right-hand drive vehicles



E146995



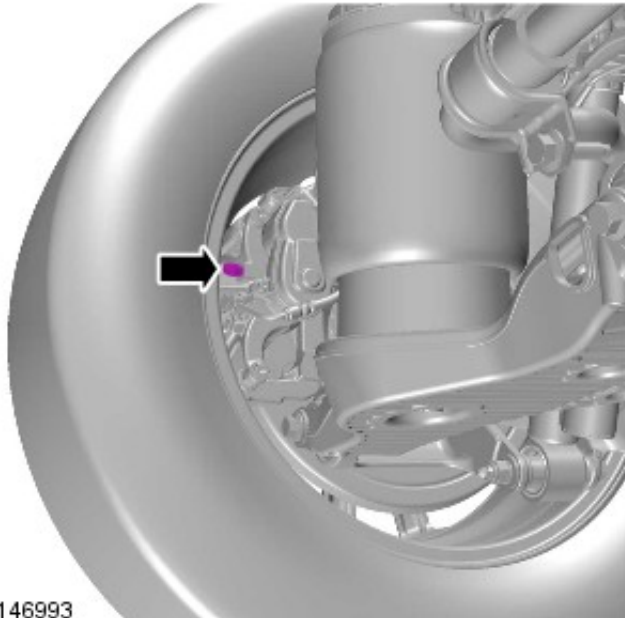
All vehicles

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14. At the end of the bleed process, depress and hold the brake pedal down.

15.

CAUTION:

Make sure the bleed screw cap is installed after bleeding.
This will prevent corrosion to the bleed screw.



E146993

With the brake pedal fully depressed, tighten the bleed screw.

Torque: 11 Nm

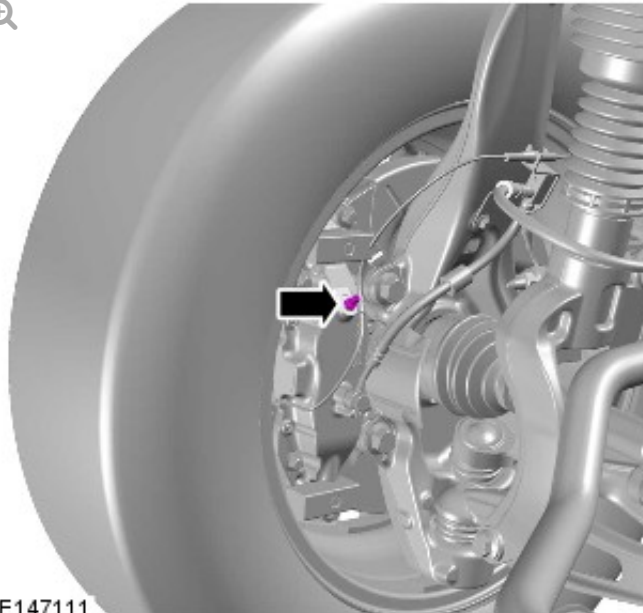


Vehicles with standard brakes

16.

CAUTION:

Make sure the bleed screw cap is installed after bleeding.
This will prevent corrosion to the bleed screw.



With the brake pedal fully depressed, tighten the bleed screw.

Torque: 8 Nm



Vehicles with high performance brakes

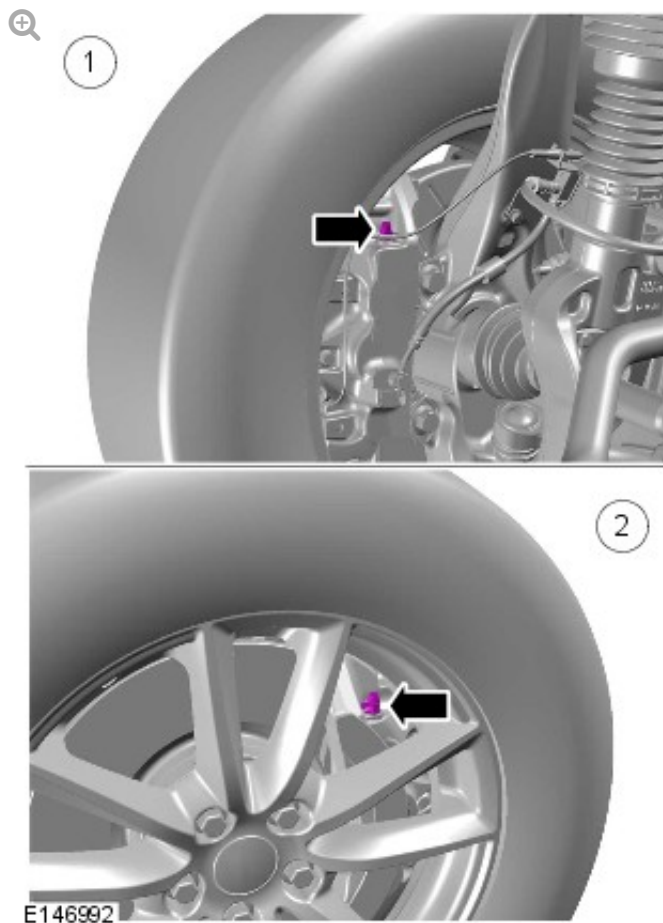
17.

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High performance front brake calipers have two bleed points. The inner bleed point must be bled before the outer bleed point. Braking efficiency may be seriously impaired if an incorrect bleed sequence is used.

CAUTION:

Make sure the bleed screw cap is installed after bleeding. This will prevent corrosion to the bleed screw.



With the brake pedal fully depressed, tighten the bleed screw.

Torque: 8 Nm



All vehicles

18. Fill the brake fluid reservoir to the MAX mark.

19.

WARNING:

Braking efficiency may be seriously impaired if an incorrect bleed sequence is used.

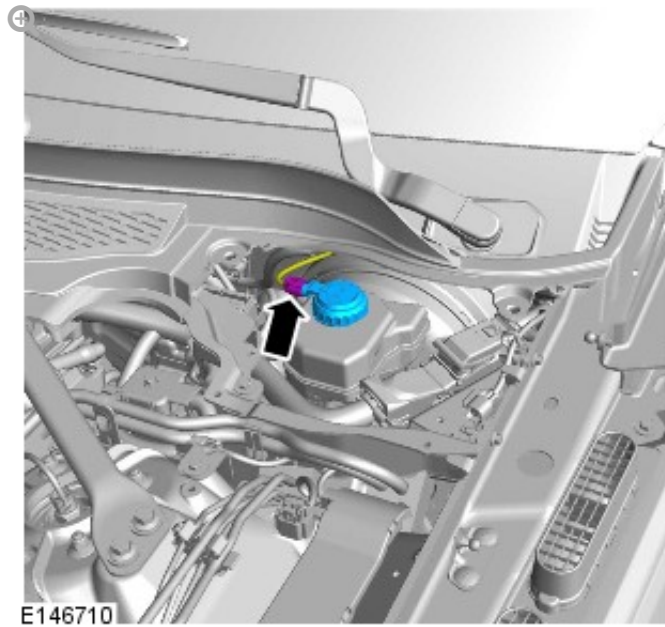
Repeat the brake bleeding procedure for each brake caliper, following the above sequence.

20. Apply the brakes and check for leaks.

21.

NOTE:

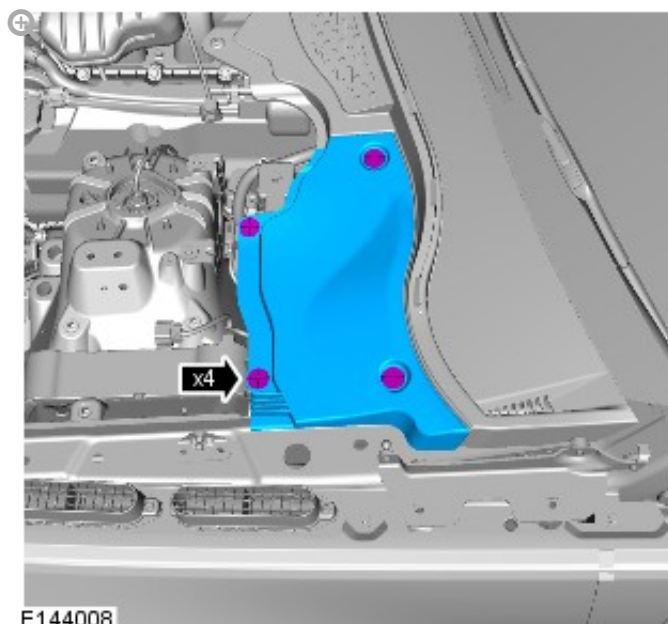
LHD illustration shown, RHD is similar.



22.

NOTE:

LHD illustration shown, RHD is similar.



E144008